

Daily Problem Solutions Booklet

Comprehensive Step-by-Step Analytical Solutions

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Integrals

1.1 Medium Integral

Inverse Hyperbolic Tangent Integral

Evaluate the definite integral:

$$\int_{-\infty}^{\infty} \tanh^{-1} \left(\frac{1}{1+x^2} \right) dx$$

Solution: Let I be the integral. Since the integrand is an even function, we can evaluate it over $[0, \infty)$ and multiply by 2:

$$I = 2 \int_0^{\infty} \tanh^{-1} \left(\frac{1}{1+x^2} \right) dx$$

We apply the logarithmic definition of the inverse hyperbolic tangent, $\tanh^{-1}(z) = \frac{1}{2} \ln \left(\frac{1+z}{1-z} \right)$:

$$\tanh^{-1} \left(\frac{1}{1+x^2} \right) = \frac{1}{2} \ln \left(\frac{1 + \frac{1}{1+x^2}}{1 - \frac{1}{1+x^2}} \right) = \frac{1}{2} \ln \left(\frac{\frac{x^2+2}{x^2+1}}{\frac{x^2}{x^2+1}} \right) = \frac{1}{2} \ln \left(\frac{x^2+2}{x^2} \right)$$

Substituting this back into the integral:

$$I = 2 \int_0^{\infty} \frac{1}{2} \ln \left(1 + \frac{2}{x^2} \right) dx = \int_0^{\infty} \ln \left(1 + 2x^{-2} \right) dx$$

We use integration by parts, letting $u = \ln(1 + 2x^{-2})$ and $dv = dx$. Then $du = \frac{1}{1+2x^{-2}}(-4x^{-3}) dx = \frac{-4}{x(x^2+2)} dx$, and $v = x$:

$$I = \left[x \ln \left(1 + \frac{2}{x^2} \right) \right]_0^{\infty} - \int_0^{\infty} x \left(\frac{-4}{x(x^2+2)} \right) dx$$

Evaluating the boundary terms gives 0 at both ∞ and 0. The integral simplifies to:

$$I = \int_0^{\infty} \frac{4}{x^2+2} dx = \left[\frac{4}{\sqrt{2}} \arctan \left(\frac{x}{\sqrt{2}} \right) \right]_0^{\infty}$$

$$I = 2\sqrt{2} \left(\frac{\pi}{2} - 0 \right) = \pi\sqrt{2}$$

1.2 Hard Integral

Greatest Common Divisor Step Integral

Evaluate the definite integral:

$$\int_1^{\infty} \frac{\gcd(3, \lfloor x \rfloor) - 1}{x^2} dx$$

Solution: Let $n = \lfloor x \rfloor$. Over the interval $x \in [n, n+1)$, the value $\gcd(3, n) - 1$ is a constant. We can break the integral into a sum of integrals over these intervals:

$$\int_1^{\infty} \frac{\gcd(3, \lfloor x \rfloor) - 1}{x^2} dx = \sum_{n=1}^{\infty} \int_n^{n+1} \frac{\gcd(3, n) - 1}{x^2} dx$$

Evaluating the integral of $1/x^2$ gives:

$$\sum_{n=1}^{\infty} (\gcd(3, n) - 1) \left[-\frac{1}{x} \right]_n^{n+1} = \sum_{n=1}^{\infty} (\gcd(3, n) - 1) \left(\frac{1}{n} - \frac{1}{n+1} \right)$$

Notice that $\gcd(3, n) = 3$ if n is a multiple of 3, and $\gcd(3, n) = 1$ otherwise. Therefore, $\gcd(3, n) - 1 = 2$ if $3|n$, and 0 otherwise. We can rewrite the sum strictly over multiples of 3, setting $n = 3k$:

$$2 \sum_{k=1}^{\infty} \left(\frac{1}{3k} - \frac{1}{3k+1} \right) = 2 \sum_{k=1}^{\infty} \int_0^1 (x^{3k-1} - x^{3k}) dx$$

Swapping the sum and the integral:

$$2 \int_0^1 \sum_{k=1}^{\infty} x^{3k-1} (1-x) dx = 2 \int_0^1 x^2 (1-x) \sum_{j=0}^{\infty} x^{3j} dx$$

Using the geometric series sum for $\sum_{j=0}^{\infty} x^{3j} = \frac{1}{1-x^3}$ for $|x| < 1$:

$$2 \int_0^1 \frac{x^2(1-x)}{1-x^3} dx = 2 \int_0^1 \frac{x^2(1-x)}{(1-x)(1+x+x^2)} dx = 2 \int_0^1 \frac{x^2}{x^2+x+1} dx$$

We can rewrite the integrand as $1 - \frac{x+1}{x^2+x+1}$:

$$2 \left(\int_0^1 1 dx - \int_0^1 \frac{x+1}{x^2+x+1} dx \right)$$

To integrate the rational function, split the numerator to create the derivative of the denominator ($2x+1$):

$$\begin{aligned} \int_0^1 \frac{\frac{1}{2}(2x+1) + \frac{1}{2}}{x^2+x+1} dx &= \frac{1}{2} \left[\ln(x^2+x+1) \right]_0^1 + \frac{1}{2} \int_0^1 \frac{1}{(x+1/2)^2 + 3/4} dx \\ &= \frac{1}{2} \ln(3) + \frac{1}{2} \left[\frac{2}{\sqrt{3}} \arctan \left(\frac{2x+1}{\sqrt{3}} \right) \right]_0^1 = \frac{\ln 3}{2} + \frac{1}{\sqrt{3}} \left(\frac{\pi}{3} - \frac{\pi}{6} \right) = \frac{\ln 3}{2} + \frac{\pi}{6\sqrt{3}} \end{aligned}$$

Substitute this back to get the final exact value:

$$2 \left(1 - \frac{\ln 3}{2} - \frac{\pi}{6\sqrt{3}} \right) = 2 - \ln 3 - \frac{\pi}{3\sqrt{3}}$$

Differentials

2.1 Medium Differential

Complex Contour Integral of a Non-Linear ODE

A complex function $f(z)$ satisfies:

$$f'(z) = if(z) - 2ze^{-iz}[f(z)]^2$$

Knowing that $f(z)$ has non-zero residues for $z = i$ and $z = -i$, compute $\oint_C f(z) dz$, where C is a positively oriented circle with radius of 2 centered at the origin.

Solution: The given differential equation is a Bernoulli differential equation of the form $y' - iy = -2ze^{-iz}y^2$. Let $u = [f(z)]^{-1}$. Then $u' = -[f(z)]^{-2}f'(z)$. Dividing the original equation by $[f(z)]^2$ yields:

$$-u' - iu = -2ze^{-iz} \implies u' + iu = 2ze^{-iz}$$

We find an integrating factor: $\mu(z) = e^{\int i dz} = e^{iz}$. Multiplying by the integrating factor:

$$u'e^{iz} + iue^{iz} = 2z \implies (ue^{iz})' = 2z$$

Integrating both sides with respect to z :

$$ue^{iz} = z^2 + C \implies u = (z^2 + C)e^{-iz}$$

Therefore, the function is $f(z) = \frac{e^{iz}}{z^2 + C}$. We are given that $f(z)$ has non-zero residues at $z = i$ and $z = -i$, which means these must be poles of $f(z)$. The poles occur when $z^2 + C = 0$. Since $i^2 + C = 0$ and $(-i)^2 + C = 0$, we find $C = 1$. Thus, $f(z) = \frac{e^{iz}}{z^2 + 1}$. The contour C is a circle of radius 2, which encloses both poles $z = i$ and $z = -i$. Using Cauchy's Residue Theorem:

$$\oint_C f(z) dz = 2\pi i (\text{Res}(f, i) + \text{Res}(f, -i))$$

Calculate the residues for the simple poles:

$$\text{Res}(f, i) = \lim_{z \rightarrow i} (z - i) \frac{e^{iz}}{(z - i)(z + i)} = \frac{e^{-1}}{2i}$$

$$\text{Res}(f, -i) = \lim_{z \rightarrow -i} (z + i) \frac{e^{iz}}{(z - i)(z + i)} = \frac{e^1}{-2i}$$

Summing the residues and multiplying by $2\pi i$:

$$\oint_C f(z) dz = 2\pi i \left(\frac{1}{2ie} - \frac{e}{2i} \right) = \pi \left(\frac{1}{e} - e \right) = -2\pi \sinh(1)$$

2.2 Hard Differential

Functional Differential Equation

If $f : \mathbb{R} \setminus \{-1\} \rightarrow \mathbb{R}$ is a differentiable function which satisfies:

$$f(x + f(y) + xf(y)) = y + f(x) + yf(x)$$

for all $x, y \in \mathbb{R} \setminus \{-1\}$, find the value of $2010[1 + f(2009)]$.

Solution: Notice the structure of the equation can be factored. The terms $(x + f(y) + xf(y))$ can be rewritten as $x + f(y)(1 + x)$. The equation becomes:

$$f(x + f(y)(1 + x)) = y + f(x)(1 + y)$$

Let $x = 0$. The equation reduces to $f(f(y)) = y + f(0)(1 + y)$. Because the right side is linear in y , the mapping $y \mapsto f(f(y))$ is bijective (assuming $1 + f(0) \neq 0$), which implies f must also be bijective. Therefore, there exists some a such that $f(a) = 0$. Substituting $y = a$ into the original modified equation:

$$f(x + f(a)(1 + x)) = a + f(x)(1 + a) \implies f(x) = a + f(x) + af(x)$$

This simplifies to $a(1 + f(x)) = 0$. Since $f(x)$ is non-constant, a must be 0. Hence, $f(0) = 0$. Substituting $x = 0$ back in gives the involution property: $f(f(y)) = y$. Now, differentiate the original equation with respect to x :

$$f'(x + f(y)(1 + x)) \cdot (1 + f(y)) = f'(x)(1 + y)$$

Let $x = 0$. Knowing $f(0) = 0$:

$$f'(f(y))(1 + f(y)) = f'(0)(1 + y)$$

Differentiate $f(f(y)) = y$ with respect to y to get $f'(f(y))f'(y) = 1$. Let $y = 0$ to find $f'(0)^2 = 1 \implies f'(0) = \pm 1$. If $f'(0) = 1$, we get $f'(f(y))(1 + f(y)) = 1 + y$. Multiplying by $f'(y)$ gives $1 + f(y) = f'(y)(1 + y)$, which solves to $f(y) = y$. However, this function lacks a singularity at $x = -1$, making it unnatural for the prescribed restricted domain. If $f'(0) = -1$, we get $f'(f(y))(1 + f(y)) = -(1 + y)$. Multiplying by $f'(y)$ gives $1 + f(y) = -f'(y)(1 + y)$.

$$\frac{f'(y)}{1 + f(y)} = \frac{-1}{1 + y}$$

Integrating both sides yields $\ln |1 + f(y)| = -\ln |1 + y| + C$. Using $f(0) = 0 \implies C = 0$.

$$1 + f(y) = \frac{1}{1 + y} \implies f(y) = \frac{-y}{1 + y}$$

This function is perfectly matched to the domain $\mathbb{R} \setminus \{-1\}$. We must evaluate $2010[1 + f(2009)]$. Knowing $1 + f(x) = \frac{1}{1+x}$:

$$2010 \left[\frac{1}{1 + 2009} \right] = 2010 \left[\frac{1}{2010} \right] = 1$$

Derivatives

3.1 Medium Derivative

Implicit Algebraic Derivative Limit

Let $(f(x) - 2)^2 = xf(x) - 2x^2$. Find $\lim_{x \rightarrow 1} f'(x)$ if $f'(x) \geq 0$.

Solution: We evaluate the equation at $x = 1$ to find the corresponding values for $f(1)$:

$$(f(1) - 2)^2 = 1 \cdot f(1) - 2(1)^2 \implies (f(1))^2 - 4f(1) + 4 = f(1) - 2$$

$$(f(1))^2 - 5f(1) + 6 = 0 \implies (f(1) - 2)(f(1) - 3) = 0$$

Thus, $f(1) = 2$ or $f(1) = 3$. Next, we implicitly differentiate the original equation with respect to x :

$$2(f(x) - 2)f'(x) = f(x) + xf'(x) - 4x$$

We evaluate this derivative at $x = 1$ for both possible values of $f(1)$. Case 1: $f(1) = 2$

$$2(2 - 2)f'(1) = 2 + 1 \cdot f'(1) - 4(1) \implies 0 = f'(1) - 2 \implies f'(1) = 2$$

This satisfies the condition $f'(x) \geq 0$. Case 2: $f(1) = 3$

$$2(3 - 2)f'(1) = 3 + 1 \cdot f'(1) - 4(1) \implies 2f'(1) = f'(1) - 1 \implies f'(1) = -1$$

This violates the condition $f'(x) \geq 0$. Therefore, the only valid branch dictates that $\lim_{x \rightarrow 1} f'(x) = 2$.

3.2 Hard Derivative

Nested Trigonometric Derivative

Let $f(x) = 1 + \sin(xf(x)^2 + x^2f(x) + 2x)$. Find $f''(0)$.

Solution: First, evaluate the function at $x = 0$:

$$f(0) = 1 + \sin(0 + 0 + 0) = 1 + 0 = 1$$

Now, we find the first derivative $f'(x)$ using the chain rule. Let $u(x) = xf(x)^2 + x^2f(x) + 2x$.

$$f'(x) = \cos(u(x)) \cdot u'(x)$$

Where $u'(x) = f(x)^2 + 2xf(x)f'(x) + 2xf(x) + x^2f'(x) + 2$. Evaluating at $x = 0$, all terms in $u'(x)$ containing x vanish:

$$u'(0) = f(0)^2 + 0 + 0 + 0 + 2 = 1^2 + 2 = 3$$

Since $u(0) = 0$, we have $f'(0) = \cos(0) \cdot 3 = 1 \cdot 3 = 3$. Next, we find the second derivative using the product and chain rules:

$$f''(x) = -\sin(u(x)) \cdot [u'(x)]^2 + \cos(u(x)) \cdot u''(x)$$

Evaluate at $x = 0$:

$$f''(0) = -\sin(0) \cdot [u'(0)]^2 + \cos(0) \cdot u''(0) = 0 + 1 \cdot u''(0) = u''(0)$$

We need to compute $u''(0)$. Let's differentiate $u'(x)$:

$$u''(x) = \frac{d}{dx} [f(x)^2 + 2xf(x)f'(x) + 2xf(x) + x^2f'(x) + 2]$$

When evaluating at $x = 0$, we can ignore the differentiation of terms where x appears to a power of 2 or higher, or apply the product rule knowing that factors of x will zero out parts of the sum.

$$u''(x) = 2f(x)f'(x) + 2[f(x)f'(x) + x(f'(x)^2 + f(x)f''(x))] + 2[f(x) + xf'(x)] + [2xf'(x) + x^2f''(x)]$$

At $x = 0$:

$$u''(0) = 2f(0)f'(0) + 2[f(0)f'(0) + 0] + 2[f(0) + 0] + 0$$

$$u''(0) = 4f(0)f'(0) + 2f(0)$$

Substitute $f(0) = 1$ and $f'(0) = 3$:

$$u''(0) = 4(1)(3) + 2(1) = 12 + 2 = 14$$

Therefore, $f''(0) = 14$.

Physics & Engineering

4.1 Mechanics

Cylinder Rolling Without Slipping

A uniform solid cylinder has mass $M = 4.0 \text{ kg}$ and radius $R = 0.25 \text{ m}$. It initially slides at $v_0 = 8.4 \text{ m s}^{-1}$ without rotating. The horizontal surface has kinetic friction coefficient $\mu_k = 0.30$. Let t measure the time from the start of the motion. Take $g = 9.8 \text{ m s}^{-2}$. Find the linear speed v , in m s^{-1} , when the cylinder first begins to roll without slipping.

Solution: While kinematic equations involving acceleration and torque can solve this, the most elegant method uses the conservation of angular momentum. Consider an axis lying on the horizontal surface along the path of the cylinder. The force of kinetic friction acts along this axis, and both gravity and the normal force are parallel/intersecting the plane, meaning their net torque about any point on this axis is zero. Therefore, the angular momentum of the cylinder about any point on the ground is conserved throughout the slipping phase. Initial angular momentum (pure translation):

$$L_i = Mv_0R$$

Final angular momentum (rolling without slipping, $v = R\omega$):

$$L_f = I_{cm}\omega + MvR$$

For a solid cylinder, $I_{cm} = \frac{1}{2}MR^2$. Since $\omega = \frac{v}{R}$:

$$L_f = \left(\frac{1}{2}MR^2\right)\left(\frac{v}{R}\right) + MvR = \frac{1}{2}MvR + MvR = \frac{3}{2}MvR$$

By conservation of angular momentum $L_i = L_f$:

$$Mv_0R = \frac{3}{2}MvR \implies v = \frac{2}{3}v_0$$

This remarkable result is independent of the mass M , radius R , friction μ_k , and gravity g . Substituting $v_0 = 8.4 \text{ m s}^{-1}$:

$$v = \frac{2}{3}(8.4) = 2 \times 2.8 = 5.6 \text{ m s}^{-1}$$

4.2 Quantum Mechanics

Photoelectric Work Function

Light of wavelength $\lambda_1 = 300 \text{ nm}$ produces a stopping potential of $V_1 = 2.50 \text{ V}$. Light of wavelength $\lambda_2 = 500 \text{ nm}$ produces a stopping potential of $V_2 = 0.90 \text{ V}$. Using only these measurements, and treating the data as exact, find the metal's

work function ϕ in electron volts.

Solution: According to the photoelectric effect equation, $E_k = \frac{hc}{\lambda} - \phi$, where E_k is the maximum kinetic energy of emitted electrons. The stopping potential V relates to this energy by $E_k = eV$. In electron volts (eV), this can be written as:

$$V = \frac{hc/e}{\lambda} - \Phi$$

Where $\Phi = \phi/e$ is the numerical value of the work function in eV. We can set up a system of linear equations from the two data points: 1) $2.50 = \frac{hc/e}{300} - \Phi$ 2) $0.90 = \frac{hc/e}{500} - \Phi$ Subtract the second equation from the first to eliminate Φ :

$$2.50 - 0.90 = \frac{hc}{e} \left(\frac{1}{300} - \frac{1}{500} \right)$$

$$1.60 = \frac{hc}{e} \left(\frac{5-3}{1500} \right) = \frac{hc}{e} \left(\frac{2}{1500} \right) = \frac{hc/e}{750}$$

Solving for the fundamental constant grouping $\frac{hc}{e}$:

$$\frac{hc}{e} = 1.60 \times 750 = 1200 \text{ eV nm}$$

Now substitute this back into either of the original equations to find Φ :

$$2.50 = \frac{1200}{300} - \Phi = 4.00 - \Phi$$

$$\Phi = 4.00 - 2.50 = 1.50 \text{ eV}$$

The metal's work function is exactly 1.50 eV.

4.3 Electric Fields

Electrostatic Energy of a Triangle

Three point charges occupy the vertices of an equilateral triangle of altitude $h = 0.45 \text{ m}$. The charges are $q_1 = 3.00 \mu\text{C}$, $q_2 = -8.00 \mu\text{C}$, and $q_3 = 11.00 \mu\text{C}$. Find the total electrostatic potential energy U of the system in joules. Take $k = 9.00 \times 10^9 \text{ N m}^2 \text{ C}^{-2}$. Give your answer to two decimal places.

Solution: First, we find the side length a of the equilateral triangle from its altitude h . For an equilateral triangle, $h = a \frac{\sqrt{3}}{2}$.

$$a = \frac{2h}{\sqrt{3}} = \frac{2(0.45)}{\sqrt{3}} = \frac{0.90}{\sqrt{3}} = 0.3\sqrt{3} \text{ m}$$

The total electrostatic potential energy U for a system of three point charges is the sum of the potential energies of each unique pair:

$$U = \frac{k}{a} (q_1 q_2 + q_2 q_3 + q_3 q_1)$$

Let's compute the sum of the charge products in $(\mu\text{C})^2 = 10^{-12} \text{C}^2$:

$$q_1q_2 = (3.00)(-8.00) = -24.00$$

$$q_2q_3 = (-8.00)(11.00) = -88.00$$

$$q_3q_1 = (11.00)(3.00) = 33.00$$

Sum = $-24.00 - 88.00 + 33.00 = -79.00 \times 10^{-12} \text{C}^2$ Now we substitute k , a , and the charge sum into the energy equation:

$$U = \frac{9.00 \times 10^9}{0.3\sqrt{3}}(-79.00 \times 10^{-12})$$

$$U = \frac{30 \times 10^9}{\sqrt{3}}(-79.00 \times 10^{-12}) = \frac{30}{\sqrt{3}}(-0.079)$$

$$U = 10\sqrt{3}(-0.079) = -0.79\sqrt{3}$$

Using the approximation $\sqrt{3} \approx 1.73205$:

$$U \approx -0.79 \times 1.73205 \approx -1.3683 \text{J}$$

Rounding to two decimal places, the total electrostatic potential energy is -1.37J .